

Double Inequalities

AQA · KS4 Year 9–10

Name: _____ Class: _____ Date: _____

Solve double (three-part) inequalities such as $a < bx + c \leq d$. Keep all three parts balanced, and list the integer solutions or show them on a number line.

VISUAL EXAMPLE



$x < 5$ on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

Only doing the operation to two parts — you must do it to ALL three parts.

Forgetting to reverse BOTH inequality signs when dividing all parts by a negative.

Mixing up which end value is included (watch $<$ versus $\leq$).

Listing integers wrongly at the ends — check whether each endpoint is included.

METHOD

- 1 A double inequality has three parts joined by two signs; treat the whole thing together.
- 2 Undo any $+$ or $-$ by doing it to all three parts.
- 3 Undo any $\times$ or $\div$ by doing it to all three parts (reverse both signs if dividing by a negative).
- 4 Write the solution as a single statement, e.g. $-1 < x \leq 3$.
- 5 If asked, list the integer solutions or draw the solution on a number line.

WORKED EXAMPLE

Solve: $1 < 2x + 3 \leq 9$

Step 1 — Three parts: 1, $2x + 3$, 9.

Step 2 — Subtract 3 from all parts: $-2 < 2x \leq 6$.

Step 3 — Divide all parts by 2 (positive, no flip): $-1 < x \leq 3$.

Step 4 — Solution: $-1 < x \leq 3$.

Step 5 — Integers: 0, 1, 2, 3 (-1 is excluded, 3 is included).

1 Solve.

(2 marks)

a) $2 < x + 1 < 5$

b) $0 < x - 2 < 4$

HOW TO START

Do the same to all THREE parts.

For $2 < x + 1 < 5$, subtract 1 from each part: $1 < x < 4$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo $\times/\div$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x < b$ | 5 List integers / draw the line if asked

2 Solve.

(2 marks)

a) $4 < 2x < 10$

b) $6 \leq 3x \leq 18$

HOW TO START

Divide all three parts by the number in front of x .

For $4 < 2x < 10$, divide each part by 2: $2 < x < 5$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Three parts — work on all of them | 2 Undo $\times/\div$ across all three | 3 Undo $\times/\div$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x < b$ | 5 List integers / draw the line if asked

3 Solve.

(3 marks)

a) $1 < 2x + 1 \leq 7$

b) $-3 \leq 2x - 1 < 5$

c) $5 \leq 3x + 2 \leq 14$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo $\times/\div$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x \leq b$ | 5 List integers / draw the line if asked

4 List the integer values of x that satisfy each inequality.

(3 marks)

a) $-2 < x \leq 3$

b) $0 \leq x < 4$

c) $-3 \leq x \leq 1$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Three parts — work on all of them | 2 Undo $+/-$ across all three | 3 Undo $\times/\div$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x \leq b$ | 5 List integers / draw the line if asked

5 Solve.

(3 marks)

a) $7 < 2x + 1 \leq 15$

b) $-4 \leq 3x - 1 < 11$

c) $2 \leq (x \div 2) + 1 < 5$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo $\times/\div$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x \leq b$ | 5 List integers / draw the line if asked

6 Solve. Take care — you will need to reverse the signs.

(3 marks)

a) $1 \quad 5 - x < 4$

b) $-2 < 3 - x \quad 2$

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Three parts — work on all of them | 2 Undo +/- across all three | 3 Undo $\times/\div$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x \leq b$ | 5 List integers / draw the line if asked

Solve 4 $2x - 6 < 10$.

(4 marks)

- Write the solution as a single inequality. (2 marks)
- List the integer values of x . (1 mark)
- Show the solution on a number line. (1 mark)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Three parts — work on all of them | 2 Undo $\pm$ across all three | 3 Undo $\times/\div$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x$ b | 5 List integers / draw the line if asked

8 a) A length L is measured as 50 cm to the nearest 10 cm. Write a double inequality (the error interval) for L . (2 marks) (5 marks)

b) Solve $3(y + 1) \div 2 < 6$ and list the integer values of y . (3 marks)

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Three parts — work on all of them | 2 Undo $\div$ across all three | 3 Undo $\times$ across all three (flip both if $\div$ negative) | 4 Write as one statement $a < x < b$ | 5 List integers / draw the line if asked

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

Why must I do every step to all three parts?

One thing I will practise before the next lesson:

Teacher key

Q1: a) $1 < x < 4$ b) $2 \leq x \leq 6$

Q2: a) $2 < x < 5$ b) $2 \leq x \leq 6$

Q3: a) $0 < x \leq 3$ b) $-1 \leq x < 3$ c) $1 \leq x \leq 4$

Q4: a) -1, 0, 1, 2, 3 b) 0, 1, 2, 3 c) -3, -2, -1, 0, 1

Q5: a) $6 < 2x \leq 14$ $3 < x \leq 7$ b) $-3 \leq 3x < 12$ $-1 \leq x < 4$ c) $1 \leq x \div 2 < 4$ $2 \leq x < 8$

Q6: a) subtract 5: $-4 \leq -x < -1$; $\times(-1)$ and reverse: $1 < x \leq 4$ b) subtract 3: $-5 < -x \leq -1$; $\times(-1)$ and reverse: $1 \leq x < 5$

Q7: a) add 6: $10 \leq 2x < 16$ $5 \leq x < 8$ b) 5, 6, 7 c) filled circle at 5, open circle at 8, shaded between

Q8: a) 45 $L < 55$ b) $\times 2$: $6 \leq y + 1 < 12$ $-1 \leq 5 \leq y < 11$ integers 5, 6, 7, 8, 9, 10
